

Cell Design Specification — VBF-T1R1-DS-01

Case: t1_r1 | Date: 2026-08-25 | Prepared: _____ | Reviewed: _____ | Approved: _____

Objective (task text, verbatim thresholds): energy density ≥ 392.61 Wh/kg; 4C fast charge without lithium plating; maximum temperature ≤ 60 °C (333.15 K); overcharge to 4.7 V without thermal runaway.

1. Basic Specification

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Electrochemical system	NMC811 / graphite (LiNi0.8Mn0.1Co0.1O2 cathode, graphite anode)	Chen2020 parameter set (task names no electrode system -> default per anchor table; funnel log entry)
Nominal capacity	5.0 Ah (parameter nominal); 5.6505 Ah simulated 1C	Chen2020 "Nominal cell capacity [A.h]"; cell/r3_l_1c.json:capacity_ah
Voltage window	2.5 – 4.2 V	Chen2020 "Lower voltage cut-off [V]" / "Upper voltage cut-off [V]"
Nominal (midpoint) voltage	4.0067 V	cell/r3_l_energy.json:midpoint_voltage_v (calc-energy mechanical derivation)
Cell dimensions	Electrode sheet 65 mm × 1580 mm; stack thickness 214.6 μm ; shell dimensions Not provided (no parameter)	Chen2020 "Electrode height [m]" 0.065, "Electrode width [m]" 1.58; cell/r3_l_energy.json:thickness_m
Electrolyte formulation	1M LiPF6 EC/EMC class with design overrides: conductivity 1.8 S/m (constant; estimate simplification vs temperature-dependent Nyman2008), transference number 0.55	Chen2020 electrolyte functions (defaults); design overrides params_r3_l.json "Electrolyte conductivity [S.m-1]" / "Cation transference number" (formulation DOF, estimate class per propose r2/r3)
Cation transference number	0.55 (default 0.2594)	params_r3_l.json
Thermal management	Total heat transfer coefficient 100 W/m2K (liquid cold-plate class)	params_r3_l.json "Total heat transfer coefficient [W.m-2.K-1]" (default 10.0)

2. Electrode and Separator

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Positive (NMC811)	75.6 (default)	0.335 (default)	0.665 (default)	3262 (default)	5.22 (default)	Al 8 μm (default 16)
Negative (graphite)	120 (default 85.2)	0.40 (default 0.25)	0.60 (default 0.75)	1657 (default)	2.5 (default 5.86)	Cu 5 μm (default 12)
Separator	6 (default 12)	0.47 (default)	—	397 (default)	—	—

All values: Chen2020 parameter dump (dump_params.py) for defaults; params_r3_l.json for overrides.

N/P ratio: formula caliber = (negative capacity density $\times$ thickness) $\div$ (positive capacity density $\times$ thickness) = $(33133 \times 0.60 \times 120 \mu\text{m}) / (63104 \times 0.665 \times 75.6 \mu\text{m}) = 0.752$. Usable-window caliber ~ 1.0 (anode operates over 0.860 of full theoretical capacity vs cathode 0.647 at the 2.5–4.2 V envelope: 1C discharge 5.6505 Ah vs full-theoretical anode 6.567 Ah / cathode 8.732 Ah, derived from $c_{\text{max}} \times \text{AMVF} \times F \times \text{thickness} \times \text{area}$). Plating safety is delivered by kinetics/transport (small particles, porous anode, high-conductivity high-t+ electrolyte) and verified by the DFN 4C exam (anode potential min +0.0249 V), not by N/P headroom — see Design Notes.

3. Process Design Parameters

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Positive areal density	163.994 g/m2 = 16.399 mg/cm2	thickness $\times$ (1-porosity) $\times$ density = $75.6 \mu\text{m} \times 0.665 \times 3262$	Chen2020 + params_r3_l.json
Negative areal density	119.304 g/m2 = 11.930 mg/cm2	$120 \mu\text{m} \times 0.60 \times 1657$	params_r3_l.json + Chen2020 density
Positive compaction density	2169.2 kg/m3 = 2.169 g/cm3	density $\times$ (1-porosity) $\div 1000$	Chen2020 3262×0.665
Negative compaction density	994.2 kg/m3 = 0.994 g/cm3	density $\times$ (1-porosity) $\div 1000$	Chen2020 1657×0.60
Electrolyte fill amount	9.384 g	pore volume $\times 1.2 \text{ g/cm}^3 \times \text{fill factor } 1.0$; pore volume = $(75.6 \times 0.335 + 120 \times 0.40 + 6 \times 0.47) \mu\text{m} \times 0.1027 \text{ m}^2 = 7.820 \text{ cm}^3$	literature electrolyte density 1.2 g/cm3 (estimate, annotated); geometry from parameter set

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Formation recommendation	0.1C CC to 4.2 V, 25 °C, 2 cycles	—	design recommended value; actual production-line value requires tuning (annotated)

4. Mass Breakdown (contract caliber: layer thickness × (1-porosity) × density × area; electrolyte excluded)

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Positive electrode layer	16.842	calc-energy caliber: 75.6 μm × 0.665 × 3262 × 0.1027 m2 (matches r3_l_energy.json layer breakdown)
Negative electrode layer	12.253	120 μm × 0.60 × 1657 × 0.1027 m2
Positive current collector (Al)	2.218	8 μm × 2700 × 0.1027 m2
Negative current collector (Cu)	4.601	5 μm × 8960 × 0.1027 m2
Separator	0.130	6 μm × (1-0.47) × 397 × 0.1027 m2
Total	36.044	cell/r3_l_energy.json:mass_kg = 0.0360436 kg (verified: recomputation matches to 1e-12)

With electrolyte fill (BOM caliber): +9.384 g -> 45.43 g. Electrolyte excluded from the ED denominator per contract (electrolyte_included: false).

5. Performance Verification

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Energy density	553.977 Wh/kg	>= 392.61 Wh/kg	PASS	cell/r3_l_energy.json:energy_density_wh_kg
Volumetric energy density	905.983 Wh/L	no registered criterion	informational	cell/r3_l_energy.json:energy_density_wh_l
4C fast-charge max temperature	325.655 K (52.51 °C)	<= 333.15 K	PASS (margin 7.50 K)	cell/r3_l_4c_dfn.json:T_max_K
4C fast-charge plating	plated = false (anode potential min +0.0249 V >= 0)	plated = false	PASS (margin 24.9 mV)	cell/r3_l_4c_dfn.json:anode_potential_v (DFN authoritative)
Overcharge 4.7 V	terminal voltage 4.7000 V; peak T 299.552 K	reaches 4.7 V	PASS	cell/r3_l_oc.json (voltage tail, T_max_K)
Thermal runaway after overcharge	triggered = false	triggered = false	PASS	cell/r3_l_tr.json:triggered (mcp = mass_kg × 900)
1C discharge capacity	5.6505 Ah	no registered criterion (vs parameter nominal 5.0 Ah)	PASS vs nominal	cell/r3_l_1c.json:capacity_ah

6. Design Notes (parameters changed in this case, with round rationale)

- Thin current collectors (Al 16->8 μm, Cu 12->5 μm, round 1): converts inactive mass into ED headroom (round-1 evaluate: ED 500–547 Wh/kg all pass).
- Anode 110->120 μm with porosity 0.40 and AMVF 0.60 (round 1 -> round 3): the DFN diagnosis (plan-update entry) showed 4C plating is transport/kinetic-limited charging from deep discharge, not end-of-charge saturation; porosity is the anti-plating spend. The round-2 AMVF 0.65 anode became near capacity-limiting (1C capacity dropped 5.743->5.602 Ah), so round 3 pairs the porosity increase with thickness compensation (recorded deviation from the round-2 reserve note, no-silent-drift rule).
- Graphite particles 5.86->2.5 μm (rounds 1–3): larger kinetic area and shorter solid-diffusion path at 20 A.
- Separator 12->6 μm (rounds 2–3): shorter ionic path (ultrathin ceramic-coated class).
- Electrolyte conductivity 1.8 S/m and transference number 0.55 (round 2–3, Stage-2 formulation fallback): the dominant plating levers — t+ alone moved the anode potential minimum from -0.1309 V to -0.0365 V (round-2 evaluate). Values are formulation-class estimates (constant-with-temperature simplification documented); not derived from a molecular funnel (no new molecule proposed, real_compute=false).
- Cooling h 10->100 W/m2K (rounds 1–3): closes the DFN Tmax gap (342.0 K at h=50 -> 325.7 K at h=100, round-2/3 evaluates).